Use two-sided large-sample Normal tests and intervals throughout. Use the following critical values unless a problem says otherwise:

Confidence Level	Significance Level α	Two-Sided Critical Value (z^*)
90%	0.10	1.645
95%	0.05	1.96
99%	0.01	2.576

For Problems 4-6, assume independent random samples from the two groups and independent observations within each group.

1. A random sample has

$$\bar{X} = 52, \qquad s = 10, \qquad n = 100.$$

Compute the estimated standard error of $\bar{X}$. Test

$$H_0 : \mu = 50 \qquad \text{vs.} \qquad H_1 : \mu \neq 50$$

at significance levels $\alpha = .10, .05, .01$ using the large-sample z-statistic. These correspond to 90%, 95%, and 99% confidence levels. Construct a 95% confidence interval for μ.

standard error: $SE(\bar{x}) = \frac{10}{\sqrt{100}} = 1$

test statistic: $z = \frac{\bar{x} - \mu_0}{SE(\bar{x})} = \frac{52-50}{1} = 2$

at $\alpha = 0.10$: $2 > 1.645$ → reject H_0 → significant

at $\alpha = 0.05$: $2 > 1.96$ → reject H_0 → significant

at $\alpha = 0.01$: $2 < 2.576$ → fail to reject H_0 → not significant

confidence interval: $CI = \bar{x} \pm z^*(SE) = (50.04, 53.96)$

2. Suppose

$$\hat{S} = 3.1, \qquad \widehat{SE}(\hat{S}) = 1.3.$$

Construct 90%, 95%, and 99% confidence intervals, and test the null hypothesis that $S = 0$ at the corresponding significance levels $\alpha = .10, .05, .01$. When do you reject or fail to reject the null of $S = 0$?

$\alpha = 0.10$: CI $= 3.1 \pm 1.645(1.3) = (0.962, 5.239)$ → reject null, interval does not contain 0

$\alpha = 0.05$: CI $= 3.1 \pm 1.96(1.3) = (0.552, 5.648)$ → reject null, interval does not contain 0

$\alpha = 0.01$: CI $= 3.1 \pm 2.576(1.3) = (-0.249, 6.449)$ → fail to reject null, interval contains 0

3. A health survey compares a biomarker to a benchmark value of 12. The sample has

$$\bar{X} = 11.5, \qquad s = 3, \qquad n = 36.$$

Is 12 a plausible value for the population mean? Construct a 95% confidence interval for $H_0 : \mu = 12$ versus $H_1 : \mu \neq 12$. Do you reject or fail to reject the null?

$SE(\bar{x}) = \frac{3}{\sqrt{36}} = 0.5$

$CI = 11.5 \pm 1.96(0.5) = (10.52, 12.48)$

12 is a plausible value as it lies within the CI which says we are 95% confident the mean lies between 10.52 to 12.48 → we fail to reject the null.

4. A real estate analyst compares the average price of a house in two neighborhoods. Group A:

$$\bar{X}_A = 25, \qquad s_A = 8, \qquad n_A = 100$$

Group B:

$$\bar{X}_B = 22, \qquad s_B = 7, \qquad n_B = 100.$$

Compute the two-sample Z-statistic for testing equality of means. Construct an approximate 95% confidence interval for $\mu_A - \mu_B$.

$SE(\mu_A - \mu_B) = \sqrt{8^2/100 + 7^2/100} = 1.063$

$z = \frac{25 - 22}{1.063} = 2.822$

$CI = (25 - 22) \pm 1.96(1.063) = (0.916, 5.084)$

We are 95% confident the mean difference between neighborhoods A and B is between 0.916 and 5.084

5. A hospital compares recovery time under two treatments. Treatment A:

$$\bar{X}_A = 14, \qquad s_A = 6, \qquad n_A = 81$$

Treatment B:

$$\bar{X}_B = 12, \qquad s_B = 5, \qquad n_B = 64.$$

Compute the standard error of the difference in means and the corresponding test statistic for

$$H_0 : \mu_A - \mu_B = 0.$$

$SE(\bar{X}_A - \bar{X}_B) = \sqrt{6^2/81 \quad 5^2/64} = 0.914$

$Z = \frac{14 - 12}{0.914} = 2.189$

6. A product-quality team compares defect scores across two factories. Factory A:

$$\bar{X}_A = 3.2, \qquad s_A = 1.5, \qquad n_A = 49$$

Factory B:

$$\bar{X}_B = 2.8, \qquad s_B = 1.4, \qquad n_B = 64.$$

Is the difference statistically large relative to the noise? Compute the test statistic and confidence interval for $H_0 : \mu_A - \mu_B = 0$, and interpret the result at $\alpha = .05$.

$SE(\bar{x}_A - \bar{x}_B) = \sqrt{(1.5)^2/49 + (1.4)^2/64} = 0.277$

$z = \frac{3.2-2.8}{0.277} = 1.446$

$CI = (3.2 - 2.8) \pm 1.96(0.277) = (-0.143, 0.943)$

we are 95% confident that the difference in means between factories A and B is between $(-0.143, 0.943)$. since 0 is in the CI, we fail to reject the null. this difference is not statistically large relative to the noise, since 0 is in the 95% CI.